Yanning DAI

PhD Student, King Abdullah University of Science and Technology (KAUST)

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Education

Ph.D. in Computer Science *Jun 2024 – Present*
King Abdullah University of Science and Technology (KAUST), Saudi Arabia GPA: 3.9/4.0
Advisor: Prof. Jürgen Schmidhuber

M.S in Electrical Engineering *Sep 2020 - Jul 2023*
Beihang University, China GPA: 3.8/4.0
Advisor: Prof. Shuo Gao

B.E. in Electrical Engineering *Sep 2016 - Jul 2020*
Beihang University, China GPA: 3.8/4.0

Research Interests

Self-improving Agents, Reinforcement Learning, Robotic Learning

Selected Publications

1. **Efficient Morphology–Control Co-Design via Stackelberg Proximal Policy Optimization.**
Yanning Dai*, Yuhui Wang*, Dylan R. Ashley, and Jürgen Schmidhuber.
ICLR, 2026.

2. **EvoAlloc: A Self-Evolving Resource Allocation Agent for Efficient Program Evolution.**
Yanning Dai, Yuhui Wang, Nanbo Li, Wenyi Wang, and Jürgen Schmidhuber.
arXiv:2610.12086, 2026.

3. **OASIS: Online Anchor-calibrated Self-Improvement System.**
Wenyi Wang, Firas Laakom, Li Nanbo, **Yanning Dai**, Yuhui Wang, Yifan Xu, Dmitrii Khizbullin, Piotr Piękos, Yilan Zhang, Yimeng Chen, Alfonso Amayuelas, Mingchen Zhuge, and Jürgen Schmidhuber.
EMNLP Findings, 2026.

4. **A Unified Framework for Rethinking Policy Divergence Measures in GRPO.**
Qingyuan Wu, Yuhui Wang, Simon Sinong Zhan, **Yanning Dai**, Shilong Deng, Sarra Habchi, Qi Zhu, Matthias Gallé, and Chao Huang.
arXiv:2602.05494, 2026.

5. **Wearable-Informed Generative Digital Avatars Predict Task-Conditioned Post-Stroke Locomotion.**
Yanning Dai, Chenyu Tang, Ruizhi Zhang, Wenyu Yang, Yilan Zhang, Yuhui Wang, Junliang Chen, Xuhang Chen, Ruimou Xie, Yangyue Cao, and Shuo Gao.
arXiv:2512.14329, 2026.

6. **Towards an Extremely Robust Baby Robot With Rich Interaction Ability for Advanced Machine Learning Algorithms.**
Mohannad Alhakami*, Dylan R. Ashley*, Joel Dunham*, **Yanning Dai**, Francesco Faccio, Eric Feron, and Jürgen Schmidhuber.
IROS, 2025.

7. **Tactile and Vision Perception for Intelligent Humanoids.**
Shuo Gao, **Yanning Dai**, and Arokia Nathan.
Advanced Intelligent Systems, 2022. [Cover Article].

8. **Advanced Electronics and Artificial Intelligence: Must-have Technologies toward Human Body Digital Twins.**
Yanning Dai, Jiaqi Wang, and Shuo Gao.
Advanced Intelligent Systems, 2022.

9. **Wearable Systems based Gait Monitoring and Analysis.**
Shuo Gao, Junliang Chen, **Yanning Dai**, and Boyi Hu.
Springer Nature, 2022. [Book].

10. **A Flexible Multi-Functional Smart Skin for Force, Touch Position, Proximity, and Humidity Sensing for Humanoid Robots.**
Yanning Dai and Shuo Gao.
IEEE Sensors Journal, 2021.

11. **A PVDF/Au/PEN Multifunctional Flexible Human–Machine Interface for Multidimensional Sensing and Energy Harvesting for the Internet of Things.**
Yanning Dai, Junliang Chen, Wenbin Tian, Lijun Xu, and Shuo Gao.
IEEE Sensors Journal, 2020.

12. **Multi-Functional Smart Skin for Multi-Dimensional Perception for Humanoid Robots.**
Yanning Dai and Shuo Gao.
IEEE FLEPS, 2020.

13. **A PVDF/Au/PEN Multifunctional Flexible Human–Machine Interface for 2D Position, Force, Hover, Surface Curvature Sensing and Energy Harvesting.**
Yanning Dai and Shuo Gao.
IEEE FLEPS, 2019.

*Equal contribution.

Experience

Research Intern *Jan 2024– May 2024*
Beijing Tsinghua Changung Hospital, Tsinghua University

Built data-driven world models for patient-specific recovery dynamics on large-scale clinical datasets.

Research Assistant *Jun 2023– Jan 2024*
Advanced Sensing and Intelligent Systems Lab, Beihang University

Developed reinforcement learning and generative models for personalized digital human systems and behavior generation in stroke rehabilitation.

Research Intern *Jun 2022– Jun 2023*
National Engineering Research Center for Ophthalmic Diagnosis

Modeled patient-specific temporal dynamics of myopia progression from longitudinal ophthalmic data.

Selected Honors

1. Rising Stars in AI, KASUT AI Initiative (30 Awardees Worldwide) *Apr 2024*

2. National Scholarship, Ministry of Education of China (Top 5%) *Dec 2022*

3. The 3rd Class Award of Science and Technology, China Instrument and Meter Society (National S&T Award, Rank: 2/6) *Oct 2022*

4. China Instrument and Control Society Scholarship, 1st Class Award, China Instrument and Meter Society (Top Society Award, Top 1%) *Nov 2019*

Skills

Programming: Python, C++, C, C#, HTML, Shell, Matlab

Project / Lib Experiences: PyTorch, TensorFlow, Transformers, Distributed Training (FSDP), Accelerated Inference (SGLang, vLLM), ROS, Robot Simulation (MuJoCo, Gazebo, Unity), Unitree SDK

Languages: English (Fluent), Mandarin (Native)

Teaching

School of Instrumentation and Optoelectronic Engineering, Beihang University:

Teaching Assistant, IntelliSense and Autonomous Systems *Spring 2023*
Teaching Assistant, Introduction to Smart Sensing Systems *Fall 2022*

Services

Conference Reviewer: NeurIPS, ICLR, EMNLP, ICRA, IEEE FLEPS.

Journal Reviewer: Expert Systems with Applications; IEEE Transactions on Neural Systems and Rehabilitation Engineering; IEEE Sensors Journal; IEEE Sensors Letters; Frontiers in Electronics.